

Determining Student’s Cognitive Models

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Abstract—Adaptive testing minimizes assessment burden while accurately estimating student ability by strategically selecting questions based on performance. We formulate adaptive question selection as a Partially Observable Markov Decision Process (POMDP) where hidden states represent student knowledge, actions are question selections, and observations are responses. We compare three decision-making approaches on 100 students from the Brazilian ENEM dataset: (1) one-step lookahead as a baseline using data-driven maximum likelihood observation models, (2) sparse sampling with Rasch IRT performing depth-2 planning with multiple trajectory samples, and (3) greedy information gain with 2PL IRT maximizing immediate variance reduction. Results show one-step lookahead achieved 86% accuracy using 31.8 questions (0.5s per student), sparse sampling achieved 79% accuracy using 27.8 questions (37.5s per student), and Monte Carlo achieved 30% accuracy using 21.2 questions (30.9s per student), illustrating trade-offs between computational cost, question reduction, and estimation accuracy.

Index Terms—component, formatting, style, styling, insert

I. PROBLEM STATEMENT

When testing student knowledge, asking hundreds of questions comprehensively evaluates understanding but causes excessive fatigue, degrading performance. Conversely, too few questions risk unreliable estimates. Moreover, question exposure produces learning effects—asking an easy question before a harder one increases success probability on the harder question. We seek the minimum number of appropriately-difficult questions to accurately assess students’ knowledge states for both summative and formative assessment.

We model this as a POMDP where hidden states represent student knowledge discretized into five levels $\{0, 1, 2, 3, 4\}$, actions are question selections from an item bank, and observations are binary responses probabilistically dependent on knowledge via Item Response Theory models. The reward encourages low belief variance while penalizing excessive questions.

Three uncertainty sources exist: hidden knowledge states must be inferred from observations, responses are stochastic (knowledgeable students may err, weak students may guess correctly), and question informativeness varies.

We compare three approaches differing in planning horizon and computational complexity:

One-Step Lookahead (Baseline): Myopic policy maximizing immediate utility using maximum likelihood observation probabilities from 1 million ENEM responses. Considers only the next question.

Sparse Sampling with Rasch IRT: Multi-step planning sampling $m = 3$ trajectories to depth $d = 2$ using Rasch IRT. Identifies high long-term information value questions at exponential computational cost.

Greedy Information Gain with 2PL IRT: Efficient policy maximizing expected variance reduction using 2PL IRT ($\alpha = 1.2$), evaluating only 15 nearest candidate questions.

All methods maintain belief distributions, update via Bayesian inference, and terminate when variance drops below threshold ($\tau \in [0.1, 0.15]$) or 45 questions are exhausted.

II. PRIOR WORK

A. Item Response Theory and the Rasch Model

Item Response Theory (IRT) provides a probabilistic framework for modeling examinees’ responses to test items, capturing both person ability and item characteristics as latent variables. Unlike classical test theory, which treats total scores as observed indicators of ability, IRT models responses at the item level and expresses the probability of a correct answer as a mathematical function of ability and item parameters [1]. Each examinee is represented by an ability parameter θ and each item by one or more parameters such as difficulty, discrimination, and guessing.

IRT has been widely applied to both simulated and real educational data to estimate student ability in large-scale and online learning environments. For example, Lee [2] used IRT and TrueSkill to estimate student ability and problem difficulty in Massive Open Online Courses (MOOCs), demonstrating that probabilistic models can effectively differentiate students’ mastery levels from noisy homework response data.

The Rasch model, proposed by Georg Rasch in 1960, is the simplest form of IRT, often referred to as the one-parameter logistic (1PL) model. It assumes that the probability of a correct response is a logistic function of the difference between a person’s ability (θ) and an item’s difficulty (β):

$$P(X_{ij} = 1 \mid \theta_i, \beta_j) = \frac{1}{1 + e^{-(\theta_i - \beta_j)}}$$

In this formulation, all items share the same discrimination power (fixed to 1), and no guessing parameter is included.

B. Applications of the Rasch Model in Educational Testing

The Rasch model has a long history in educational assessment. According to several researchers [3] [4], the Rasch

model offers advantages in educational measurement, in particular in mathematical skills assessment. Large-scale standardized assessments often utilize Rasch model for item banking and adaptive administration [5]. One of the key strengths of these models is their ability to increase measurement precision by matching item difficulty to the estimated ability of each examinee. When items are targeted in this way, tests can achieve the same level of reliability while using far fewer questions. Empirical studies have shown that adaptive tests can reduce test time by around 50% while maintaining comparable psychometric reliability to a fixed form [6]. Researchers have applied Rasch-based adaptive testing in various educational contexts to validate these advantages. Ebenbeck and Gebhardt [7], for example, developed an adaptive assessment for students with special needs using a Rasch-calibrated item bank. Their results showed that using the Rasch model to guide item selection helped reduce test time while actually improving test accuracy.

C. Adaptive Testing

Computerized Adaptive Testing (CAT) dynamically adjusts the difficulty and selection of test questions based on an examinee’s performance, aiming to estimate ability by giving students questions around their appropriate difficulty [8]. Traditional CAT algorithms often choose the next question whose difficulty is near the examinee’s current ability estimate – roughly a 50% chance of a correct answer – because such an item is most informative at that point [9]. This item typically has the largest Fisher information estimate [10]. By maximizing the immediate information gain, you can minimize the number of questions necessary. However, the fisher information assumes the current ability is known, and due to the high uncertainty, it may not be very informative. Therefore, decision making algorithmic approaches may be optimal. Adaptive testing is essentially a sequential decision-making problem under uncertainty. With each question, the testing algorithm must decide which item to administer next (the action) based on current evidence about the student’s ability (the state of knowledge). The true ability level is not directly observable; the system only receives noisy observations in the form of correct or incorrect answers. This naturally suggests a Partially Observable Markov Decision Process (POMDP) formulation for adaptive testing. In a POMDP model of CAT, the hidden state is the examinee’s true proficiency, and the algorithm maintains a belief state – a probability distribution over possible ability levels given all the responses so far [11]. Each question administered produces an observation (e.g. the student’s answer correctness), which updates the belief state. The goal is to choose a sequence of questions (actions) that maximizes some notion of reward – for example, achieving a confident estimate of ability with minimal questions (or equivalently minimizing estimation error or test length) [12]. Formulating CAT as a POMDP in principle allows one to compute an optimal questioning policy that balances exploration and measurement accuracy across multiple steps. The state (ability) is continuous or high-dimensional, and the space

of possible question sequences is enormous. Exact dynamic programming solutions are intractable, therefore approximate planning and online learning solutions are more practical.

III. APPROACH

A. Monte Carlo

We implemented a Monte Carlo one-step lookahead policy for adaptive question selection. Student ability was discretized into five states $S = \{0, 1, 2, 3, 4\}$. Item difficulties were estimated from empirical ENEM response rates using a logit transform and rescaled to $[0, 4]$. We modeled observations using a Rasch-style IRT model

$$P(\text{correct} \mid s, a) = \frac{1}{1 + \exp(-1.6(s - d(a)))},$$

with probabilities clipped to $[0.10, 0.90]$ for numerical stability.

For each student, we treated all questions they answered as eligible actions and initialized the belief $b(s)$ with a uniform prior over the five ability states, reflecting no prior information about the student. The Monte Carlo policy estimates the value of each candidate question by sampling. For each action a , we draw N samples by: (1) sampling a latent state $s \sim b(s)$, (2) sampling a hypothetical response z from the IRT model, (3) updating the belief $b'(s)$ via Bayes’ rule, and (4) computing reward $R(b') = -\text{Var}(b')$. The expected utility is

$$U_{\text{MC}}(a) = \frac{1}{N} \sum_{i=1}^N R(b'_i),$$

and the question maximizing $U_{\text{MC}}(a)$ is selected. After selection, the belief is updated using the student’s *actual* observed response from the dataset. The process repeats until variance < 0.10 , no questions remain, or 45 questions are asked.

We evaluated on 100 students and measured performance by discrete state accuracy, mean absolute error, and number of questions required to reach the stopping criterion.

B. Sparse Sampling

We developed an adaptive testing system based on sparse sampling online planning for estimating student ability using Item Response Theory (IRT). Given the objectives of the current project and the literature review, the Rasch model is an appropriate choice. The test design involves sparse sampling, where each student sees only a subset of items, and Rasch is well-suited to this scenario since it ensures that all item difficulties and student abilities can be placed on a unified scale despite each examinee answering different questions.

Our approach models student ability as a discrete belief distribution over five ability levels $\{-2, -1, 0, 1, 2\}$. The system employs the Rasch IRT model to compute the probability of a correct response as $P(\text{correct} \mid \theta, \beta) = \frac{1}{1 + \exp(-(\theta - \beta))}$, where θ represents student ability and β represents question difficulty. Question difficulties were estimated from historical response data using maximum likelihood estimation.

Our decision-making framework formulates the testing process as a Markov Decision Process (MDP) where the state is

a categorical belief distribution over the five discrete ability levels, and actions correspond to selecting which question to ask next. The transition model samples a student's true ability from the current belief distribution, simulates their response using the Rasch model, and updates the belief via Bayesian inference. The reward function is defined as $R = \text{Var}(b) - \text{Var}(b') - c$, where $\text{Var}(b)$ represents belief variance and $c = 1.0$ is a question cost penalty. For action selection, at each decision point, the algorithm evaluates each available question by sampling m possible futures to depth d , recursively estimating the expected cumulative reward. It will terminate when belief variance drops below $\tau = 0.1$ or 45 questions are asked.

Below is the rasch sparse sampling algorithm:

- Initialize a uniform belief $b(s)$, estimate item difficulties β_a , and set stopping criteria ($\tau = 0.1, 45$).
- For each test step, and for each available item a , compute its sparse-sampling value estimate.
- For each of the m samples:
 - Sample a s according to the current belief $b(s)$.
 - Compute $O(s, a) = \sigma(\theta_s - \beta_a)$.
 - Sample a response $r \in \{0, 1\}$ from this probability.
 - Update the posterior belief using:

$$b'(s) \propto \begin{cases} O(s, a) b(s), & r = 1, \\ (1 - O(s, a)) b(s), & r = 0. \end{cases}$$
 - Compute the reward as the decrease in variance minus the question cost.
 - Estimate future value by recursively evaluating the belief for depth $d - 1$.
- Average the rollout values to obtain $Q(a)$ for item a , and select the next item a^* that maximizes $Q(a)$.
- Present item a^* to the student and observe the true response, and update the belief. Remove item a^* from the available pool.
- Continue until belief variance drops below τ , items run out, or 45 questions have been asked.

C. One Step LookAhead

For our baseline, we used a one step lookahead [13] approach, where all of our parameters were learned by the data. We binned each state space into 5 states, with 1 being the lowest ability, and 5 as the highest. Difficulty was also binned this way, with 5 being the most difficult questions, and 1 being the easiest questions. The difficulty of the question were calculated based on the percentage of students who got the question correct. Using 1 million student responses from the ENEM dataset [14], we constructed a maximum likelihood [13] model $O(s, a)$ for a student answering a question correctly given the question difficulty a and student ability s . This gave us our observation function. Lastly, our reward $R(b, a)$ was determined by the negative variance of our belief, where a lower variance provides a higher reward. We assumed students could not change their ability throughout the test.

We then performed our algorithm as follows:

- For each difficulty a , find $O(s, a)$ at each state
- Obtain two update the beliefs, where $b'_1(s) \propto O(s, a) * b(s)$, and $b'_2(s) \propto (1 - O(s, a)) * b(s)$.
- Find the reward $R(b'(s))$ for those two respective beliefs.
- Find the probability ρ of getting that reward, given by $\sum O(s, a) * b(s)$
- Calculate the utility u of those beliefs where $u = \rho * r(b'_1(s)) + (1 - \rho) * r(b'_2(s))$
- Choose the difficulty that maximizes the utility.
- Give a random question of that difficulty to the student and observe the true response
- Update the belief given the true observation where $b'(s) \propto (1 - O(s, a)) * b(s)$
- Repeat until the variance is less than .1 or there are no more questions (maximum of 45 questions).

IV. ANALYSIS AND RESULTS

All methods were tested with a random selection of 100 students from the ENEM dataset [14]. The accuracy (determined by the highest belief state probability matching the true state probability from the student), number of questions per student to achieve that belief, and runtime needed to select all question before achieving a variance less than .1 is detailed below.

A. Monte Carlo

The Monte Carlo one-step lookahead policy samples hypothetical latent states and responses to approximate expected reduction in uncertainty, but it remains myopic and does not plan beyond a single question. The approach relies on a discrete five-state ability representation, which introduces quantization error, and uses a Rasch-style observation model with a fixed discrimination parameter rather than calibrated item-specific values. Moreover, per-student train/test splits reduce the available action set, and the model assumes no learning occurs during testing.

Future work should incorporate multi-step lookahead via sparse sampling or Monte Carlo tree search, estimate full IRT parameters from larger datasets, and explore continuous-state representations (e.g., particle filters). Real-world validation in classroom settings remains an open direction.

B. Sparse Sampling

We tested the system on a dataset of 100 students who each answered 45 questions (4,500 responses total). The system was configured with sparse sampling parameters depth $d = 2$ and sample size $m = 3$. Overall, the adaptive system achieved 79.00% classification accuracy in estimating student ability levels, and it required an average of only 27.81 questions per student versus the full 45-question assessment. The approach successfully converged for all students (100% reaching variance threshold) However, the computational cost of our approach was substantial, with an average runtime of 37.5s per student. This high computational demand is from the exponential growth of sparse sampling's search tree.

Model	One Step Lookahead	Sparse Sampling	Monte Carlo
Accuracy	86%	79%	30%
Questions	32	28	21
Runtime (s)	0.5	38	31

TABLE I
COMPARISON OF ADAPTIVE TESTING METHODS

C. One Step Lookahead

After comparing the most likely ability state from $b'(s)$ and comparing it to the ground truth for that student from the collected data. Using a random selection of 100 students from the ENEM dataset [14], we obtained an accuracy of 86%. On average, the model required 31.8 questions per student, and took .5 seconds to select these questions for each student.

V. DISCUSSION

As shown in Table I, our baseline one step lookahead had the best accuracy and the shortest runtime. This could be due to using the maximum likelihood instead of the Rasch model for the observation model. While the Rasch model does provide domain knowledge of the true probability of an observation and should have increased accuracy, it assumes a continuous distribution of ability. Additionally, the Rasch model assumes an item difficulty that was only estimated for our analysis, as the true difficulty depends not only on the percentage of students who answered the question correctly, but also the ability of those students. Therefore, we were most likely able to increase our accuracy by simply creating our own observation model from data.

The information gain approach achieved the lowest accuracy (50%) despite using the fewest questions (18.9 average). Several factors contributed to this poor performance. First, the 2PL IRT observation model used a fixed discrimination parameter ($\alpha = 1.2$) rather than item-specific values estimated from data, causing the model to misestimate response probabilities across different items. Second, item difficulties were estimated from empirical response rates using a simple logit transformation and linear scaling rather than proper IRT calibration that jointly estimates item parameters and student abilities. Third, the train-test split used separate items for ability estimation and testing, reducing the quality of ground truth ability estimates compared to using all available data. Most critically, the discrete five-state representation introduced substantial quantization error—students with true abilities near state boundaries (e.g., 2.8 vs 3.2) were frequently misclassified despite small actual differences. The greedy variance reduction criterion, while computationally efficient, also proved too myopic, selecting questions that reduced immediate uncertainty without considering whether those reductions would persist or be revised by subsequent questions. These compounding limitations resulted in overconfident but inaccurate belief distributions that converged quickly to incorrect states.

VI. LIMITATIONS, ASSUMPTIONS, FUTURE WORK

Our data and problem definition has a few assumptions and limitations that may limit our results for a real testing

environment. We binned the ability as state and discrete. While in most scenarios we can accurately assume student ability does not change throughout a summative assessment, ability, as well as question difficulty is continuous. Therefore, allowing for a continuous observation and belief distribution may lead to more accurate results.

More detailed limitations, assumptions, and future work for each individual policy is stated below.

A. Monte Carlo - Nicole

The Monte Carlo one-step lookahead policy evaluates only the immediate expected reduction in uncertainty and does not perform multi-step planning. This myopic behavior may lead to suboptimal early questions that delay convergence. The model employs a Rasch-style IRT observation model with a fixed discrimination parameter and item difficulties estimated from empirical response rates rather than calibrated parameters. The Monte Carlo one-step lookahead policy evaluates only the immediate expected reduction in uncertainty and does not perform multi-step planning. The model employs a Rasch-style IRT observation model with a fixed discrimination parameter and item difficulties estimated from empirical response rates rather than calibrated parameters, and it assumes no learning effects during testing.

Future work includes incorporating multi-step lookahead through sparse sampling or Monte Carlo tree search, estimating full IRT parameter sets from larger datasets, exploring continuous ability representations (e.g., particle filters), and validating the framework with real students in operational testing environments.

B. Sparse Sampling

This work relies on several simplifying assumptions that limit how broadly the method can be applied. First, we use the Rasch (1-parameter) IRT model to describe students' response behavior, which ignores important factors such as discrimination parameters and guessing parameters which present in 2PL and 3PL models. These factors are important in real testing environments, and ignoring them means the model may miss some of the nuances in how students interact with different questions. Second, ability is represented using only five discrete levels. This makes the system easier to run, but it is clearly too coarse for situations where fine-grained ability estimates matter.

The computational cost of sparse sampling is another major constraint. Because its complexity grows as $O(|A|^d \times m^d)$ complexity, the algorithm becomes infeasible for larger question banks. In our setting, the method already requires about 37.5s for a single student, which would not scale well in large-scale testing contexts.

These limitations point to several directions for future work. Moving from discrete to continuous ability representations would make ability estimation more accurate. Incorporating 2PL or 3PL IRT models would also allow the model to capture item discrimination and guessing behavior, therefore more accurate.

C. One Step Lookahead

For the One-Step Lookahead policy, we estimated the difficulty using the percent of students who answered the question correctly, and then aggregated that difficulty into 5 discrete bins. However, within a single bin, questions may vary in difficulty, which can contribute to a overall lower accuracy.

For our observation model, we were able to use a maximum likelihood model because we only had 5 bins. Each student only had a total of 45 questions. This gives an average of 9 questions per bin difficulty. While this was sufficient, increase the bin number by 10 would greatly decrease the available questions per bin, thus decreasing the accuracy of the observation model. This can be accounted for in future work by instead using a prior, most likely informed by the Rasch model.

VII. CONCLUSION

This work evaluated adaptive testing as a sequential decision-making problem formulated as a POMDP, where policies must balance question reduction, computational cost, and estimation accuracy. Through experiments on 100 ENEM students, we demonstrated that the one-step lookahead policy achieved the highest accuracy and fastest runtime, making it well-suited for real-time assessment. Sparse sampling provided a favorable reduction in the number of questions needed; however, this benefit came with significantly higher computational cost, limiting its practicality at scale. The Monte Carlo method offered computational efficiency with fewer questions but incurred lower accuracy due to its myopic one-step planning and reliance on fixed IRT parameters.

Overall, our findings highlight a fundamental trade-off between planning depth and operational feasibility. More sophisticated lookahead provides theoretical value but is constrained by runtime in live testing environments, while lightweight methods deliver fast and accurate estimates but may forego long-term gains. Future work should explore continuous-state representations, calibrated IRT models, and multi-step planning techniques such as sparse sampling or Monte Carlo tree search with pruning to strike a more effective balance between accuracy, test length, and computation. Integrating these capabilities in authentic educational settings presents a promising direction for advancing adaptive assessment.

VIII. CONTRIBUTIONS

Elizabeth Childs worked on the One-Step lookahead, literature review for adaptive learning, and discussion section.

Zhenze Zhang worked on the sparse sampling, literature review for item response theory, rasch model, and the applications of rasch model in educational testing.

Nicole Mulvey worked on the Monte Carlo simulation, and any part of the paper related to the discussion of it. In addition, she also typed the abstract, problem statement, and conclusion.

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